Computer Networks

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Course Information

- Credits/Hours: 3/48
- Course Web site:

https://lexue.bit.edu.cn/course/view.php?id=17931

100071011 Computer Networks （English） - 2023

  - Login Name: your student ID
  - Login Pass: your pw
  - Download PPT/PDF
  - Upload your homework/papers/reports

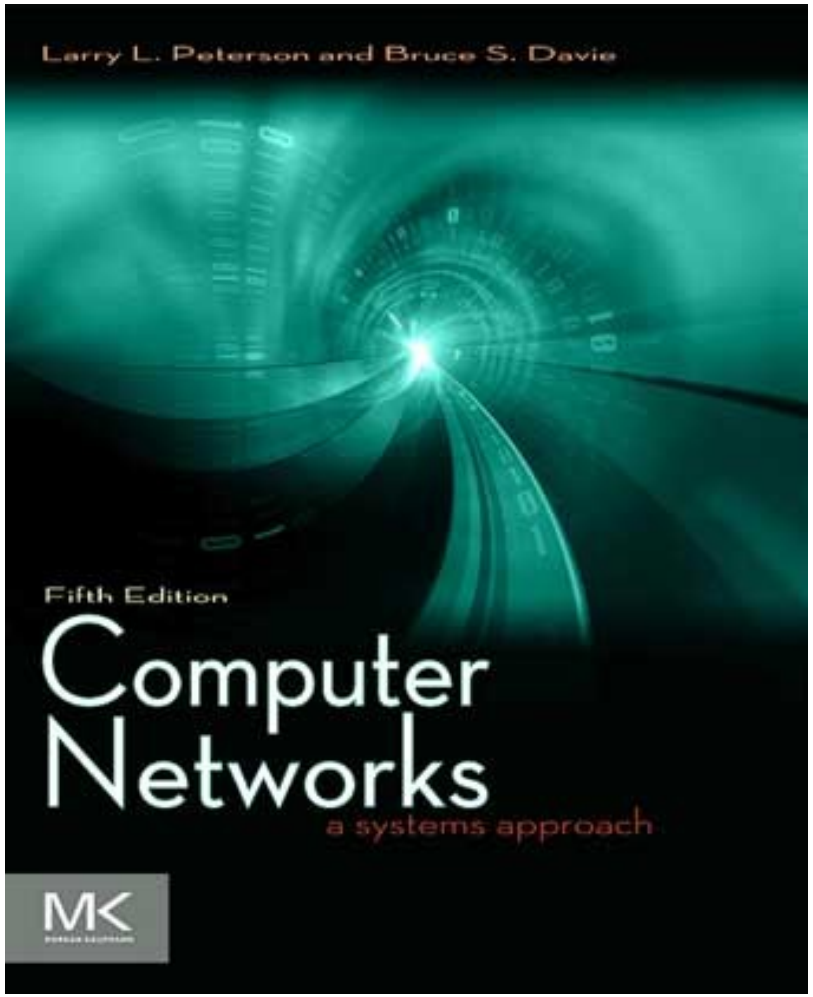
Goals of this course

- Learn and get **familiar with** the basic concepts and aspects of computer networks;
  - ✓ Switching, Forwarding, Routing, Naming, ...
- **Thoroughly understand** network architectures, primary ideas of network engineering, as well as applications;
  - ✓ Layering, ...
- **Build up ground work** for further study and R&D in computer networks field.

What You Learn in This Course

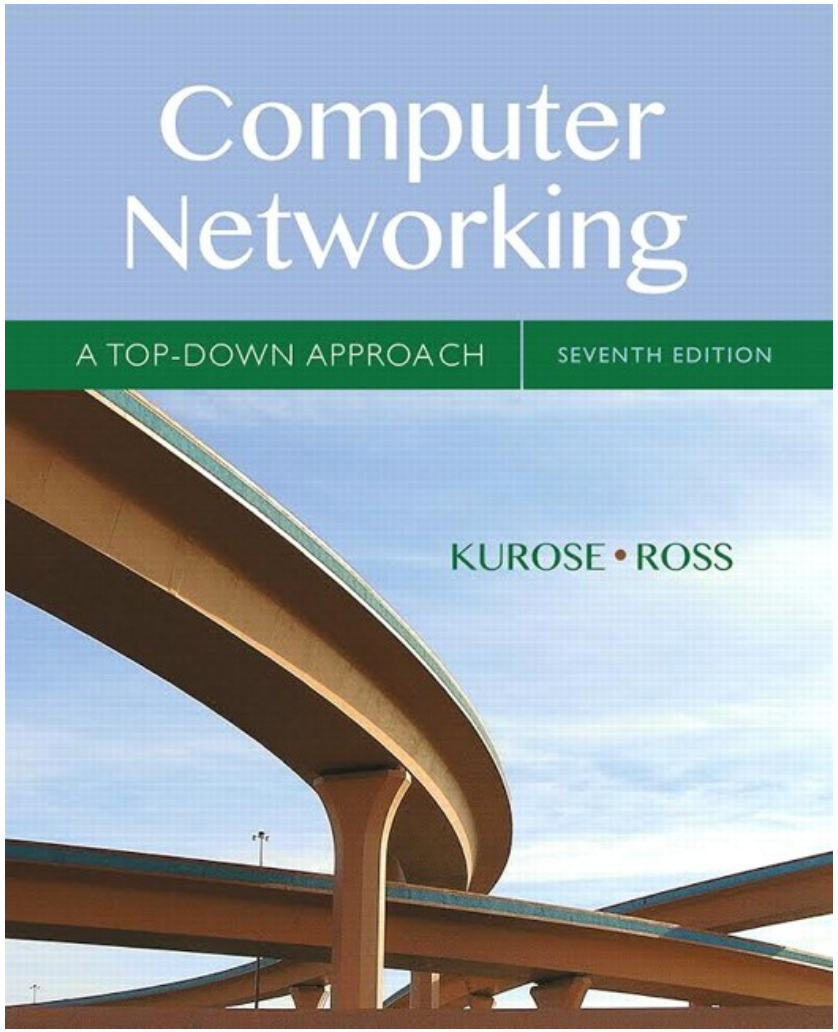
- **Knowledge:** how the network (e.g. Internet) works
  - Network (Internet) architecture
  - TCP/IP Protocol suite
  - Applications (Web, e-mail, FTP, P2P, …)
- **Insight:** key concepts in networking
  - Protocols
  - Layering
- **Skill:** network engineering/network programming
  - Network design/configuration/troubleshooting
  - Socket programming

Recommended Reference Textbooks



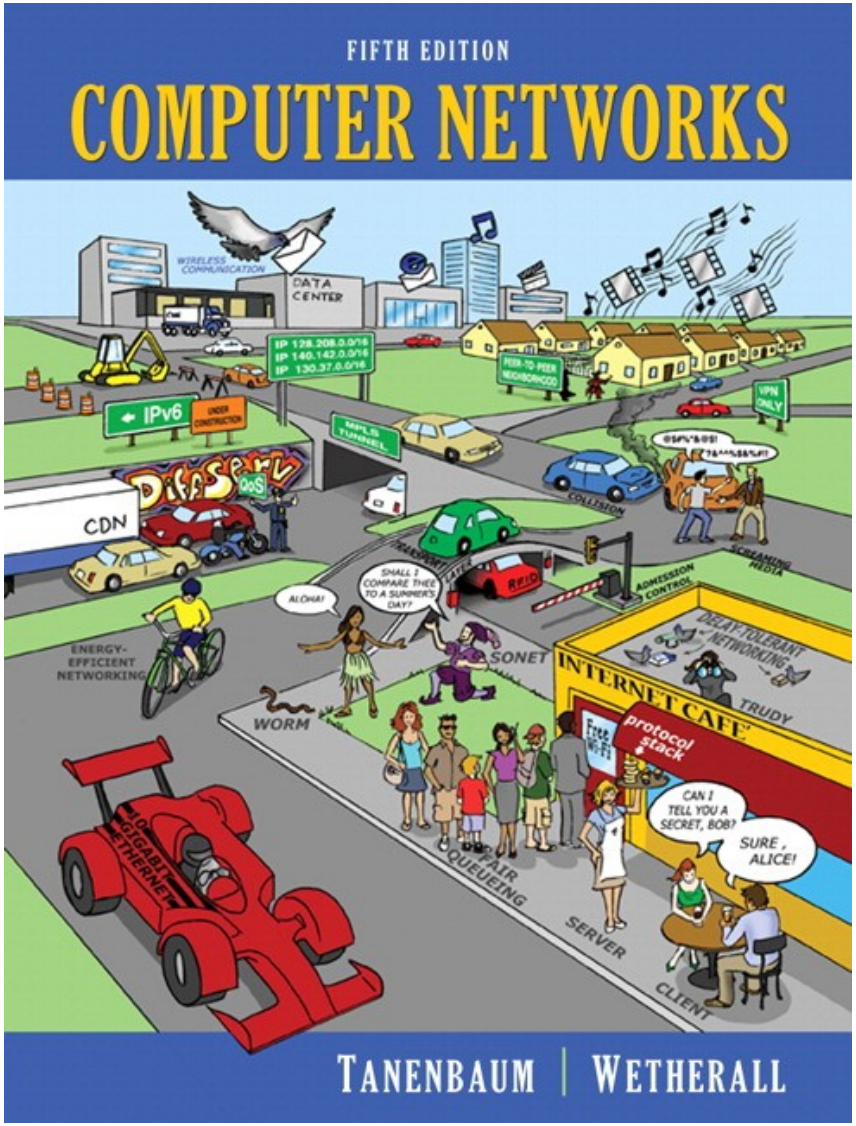
- Computer Networks: A Systems Approach (5th / 6th Edition), Larry L.Peterson, Bruce S.Davie.

Recommended Reference Textbooks



- Computer Networking: A Top-Down Approach (6th / 7th / 8th Edition), James F. Kurose, Keith W. Ross.

Recommended Reference Textbooks



- Computer Networks, 5th / 6th edition
  - Andrew Tanenbaum

Grading Policy

Final Grade	
Attendance / Written Assignments / Quiz	10%
Configuration and Analysis Experiments	10%
Programming Projects	10%
Course paper	10%
Final exam	60%

Assignments

- All assignments (homework, Networking Experiments, Programming Projects, Course paper, etc.) are individual;
- Start early;
- Ask questions early;
- Submit on time.

Focuses on the Internet

- **Internet only one that matters for computer science**
  - When computers use these other networks for IP-based communication, they are part of the Internet!
  - The Internet is not a network "technology", it is the paradigm that ties networks together! (Internet)

The Internet: an hourglass *with layers*

